

SCIENTIFIC BASIS · METHODOLOGY AND RATIONALE

Why this data, this model, and this much caution

The reasoning behind TerraPulse's prospectivity engine: the deposit model it implements, the geological control each evidence layer stands for, why those data sources and not others, and the deliberate limits on what it will claim. Written for **geologists who need to agree or disagree with a specific choice**, not be reassured by one.

1 · The design principle

A prospectivity model can be built two ways. The choice made here determines everything downstream.

The conventional machine-learning approach trains a classifier on known deposits and lets it discover its own weights. That requires a large, clean, spatially unbiased training set of drill outcomes — including the negatives. Almost no exploration company has one, and a model trained on discovered deposits learns where people have already looked, which is a map of historical access as much as of geology.

TerraPulse instead implements a **transparent expert-weighted evidence model**. Each layer is normalised to a common scale, multiplied by a published weight, and summed into a 0–100 prior. The weights encode a stated deposit model rather than being fitted to data that does not exist.

An XGBoost regressor is then fitted **to that prior**, purely so that SHAP can attribute each cell's score back to the contributing layers. The machine learning here is not discovering geology — it is making an explicit weighted sum explainable per cell. That distinction matters when someone asks what the model learned: the honest answer is that it learned to reproduce a stated expert prior, and the SHAP values decompose that prior rather than revealing a hidden pattern.

WHY IT IS BUILT THIS WAY

The fitted surrogate is the **seam where real drill outcomes replace the expert prior**. When a customer accumulates labelled results, the same pipeline trains on those instead, and every downstream surface — scores, SHAP panels, reports — keeps working unchanged. Starting with a defensible prior means the product is useful before that data exists, and upgradeable once it does.

2 · Why Birimian orogenic gold first

Four reasons, in order of weight.

01 It is where the pilot users are

Ghana and West Africa, exploring gold in Birimian greenstone belts. A model that serves the first real users is worth more than four models that serve none of them fully.

02 The mineral system is unusually well characterised

Orogenic gold has a coherent, widely published exploration model: mineralisation on second- and third-order structures splaying off crustal-scale shear zones, in metavolcanic and metasedimentary hosts, with carbonate–sericite–chlorite alteration and gold–arsenic pathfinder haloes. There is a real consensus to encode.

03 The evidence it needs is obtainable everywhere

Magnetics, soil geochemistry, satellite alteration ratios, mapped lithology and known-occurrence proximity are all either routinely collected by explorers or available as global public data. A model requiring downhole geophysics or seismic could not run on a new project's first day.

04 Depth before breadth

Building porphyry, lithium and nickel profiles before a single gold model is field-validated would multiply the surface area of things that could be quietly wrong. The architecture keeps the seam open; the discipline is to not use it yet.

3 · The weights, and what each stands for

Five evidence layers, weights summing to 1.0. Published rather than hidden — a geologist should be able to disagree with a specific number.

Geochemical anomaly	0.30	
Structure proximity	0.25	
Magnetic amplitude	0.20	
Lithology	0.15	
Deposit proximity	0.10	

Weights are **renormalised over whatever layers a project actually has**, so a project without geochemistry still scores on its remaining evidence rather than being penalised into uniformity.

Geochemical anomaly 0.30 · highest

THE GEOLOGICAL CONTROL

Gold and its pathfinder elements — arsenic in particular — form dispersion haloes in soil and regolith that are considerably wider than the mineralised structure itself. This is the most direct evidence that a mineralising system was present, which is why it carries the largest weight.

HOW IT IS COMPUTED

Each element's distribution is reduced to **85th, 95th and 99th percentile thresholds computed from the project's own sample population**. A sample's anomaly class is the highest bucket reached by any of its elements. Classes are binned to the analysis grid, each cell taking the maximum class falling within it.

WHY PERCENTILES, NOT FIXED CUT-OFFS

Absolute thresholds in ppm are meaningless across different regolith regimes, laboratory methods and detection limits. Percentile classification against the local population is standard exploration-geochemistry practice precisely because anomaly is a **relative** concept — it asks whether this sample is unusual for this survey.

KNOWN LIMITATION

Percentile thresholds assume the survey covers enough background to define one. A survey deliberately collected only over a known anomaly will classify its own centre as background.

Structure proximity 0.25

THE GEOLOGICAL CONTROL

Orogenic gold is structurally controlled. Fluid moves along faults, shears and lithological contacts, and deposits sit at the second-order structures splicing off crustal-scale breaks. Locating the plumbing is the single most transferable exploration heuristic in this deposit class.

HOW IT IS COMPUTED

The **horizontal-gradient magnitude of the magnetic field**. Contacts, faults and shears appear as edges in magnetic data, and edges are where the gradient is steep. Computed as the magnitude of the field's spatial gradient across the analysis grid.

WHY GRADIENT RATHER THAN AMPLITUDE

Gradient is **sign-invariant**. It answers "is there a discontinuity here" without needing to decide whether high or low field values are favourable — which, as the next layer explains, is the model's least certain assumption. Published Ashanti and Birimian aeromagnetic practice leans on derivatives (analytic signal, first vertical derivative, horizontal gradient) to map controlling structures, rather than on raw amplitude polarity. That is why gradient outweighs amplitude here.

Magnetic amplitude — scored inverted 0.20 · least certain

THE CLAIM

A magnetic **low** is treated as favourable. This is the only inverted layer in the model.

THE BASIS

The orogenic-gold alteration assemblage is magnetite-destructive. Carbonate alteration — calcite, dolomite, ankerite, siderite with sericite, pyrite and albite — is classed as partially magnetite-destructive with decreased susceptibility; phyllic alteration (quartz–sericite–pyrite–chlorite) as magnetite-destructive outright. The alteration is dominated by silica, sericite, carbonate and chlorite, and so rarely produces highly magnetic mineralogy.

SOURCE

"Magnetic signatures of mineralization are not always associated with strongly magnetic sources, but can be indicated by zones of anomalously weak magnetization."

Clark (2017), *Magnetic effects of alteration in mineral systems*, Exploration '17 Petrophysics Workshop

STATED LIMITATION — THE MODEL'S WEAKEST LINK

Three documented counter-cases, all of which the team names rather than hides. **Pyrrhotite is ambiguous** — monoclinic pyrrhotite is strongly magnetic and occurs both as an ore-associated sulphide and in barren sulphidic shales adjacent to lodes, so it can produce magnetic highs on or near mineralisation; pyrite dominates at low metamorphic grade and pyrrhotite at higher grade, making the correct polarity partly a function of grade. **Host rock decides whether there is a signal at all** — Ashanti Belt hosts include magnetic basalt and andesite as well as effectively non-magnetic graphitic and chlorite phyllites. **Banded-iron-formation-hosted orogenic gold is a magnetic high**, not a low.

HOW THE UNCERTAINTY IS HANDLED

The decision is recorded as a formal decision record with its rationale, counter-cases and a validation plan; it is asserted as **more defensible than the previous behaviour, not as settled**. The polarity lives in a single named constant, which is the intended configuration seam for making it per-commodity or per-host-rock. Two regression tests pin the behaviour so it cannot drift silently.

Lithology 0.15

THE GEOLOGICAL CONTROL

Birimian orogenic gold is hosted preferentially in metavolcanic and metasedimentary greenstone-belt rocks, with intrusive contacts a secondary host. Host rock is a genuine control on where a system can form.

HOW IT IS COMPUTED

Mapped geological units are sampled from Macrostrat and their lithology strings classified onto a favourability scale: **mafic metavolcanics 1.0** (basalt, andesite, amphibolite, tuff — the prime greenstone host), **metasediments 0.85** (greywacke, argillite, phyllite, schist — major host of Ashanti-belt deposits), **granitoids 0.5** (favourable at contacts, lower overall), gneiss and migmatite 0.4, carbonates 0.3, and **cover 0.15** — laterite, alluvium and regolith are penalised because they mask bedrock signal rather than host it.

WHY THIS IS SAMPLED COARSELY

Geology varies slowly relative to a 50 m analysis cell. Units are sampled on a coarse lattice and nearest-filled onto the grid — ten thousand individual queries would be absurd for a surface that changes over kilometres.

THE COMMODITY DEPENDENCE

This table is **the most gold-specific component in the model** and is close to inverted for other deposit classes: a lithium-pegmatite or porphyry-copper model wants fractionated granite and intrusives ranked high, exactly where this table ranks them low.

Deposit proximity 0.10 · lowest

THE GEOLOGICAL CONTROL

Orogenic gold clusters along greenstone belts at camp and belt scale. Proximity to a known occurrence is genuine evidence that the right crustal architecture is present.

HOW IT IS COMPUTED

Exponential decay of great-circle distance to the nearest reference deposit, with a **25 km decay constant** chosen to match belt-scale clustering rather than deposit-scale.

WHY IT CARRIES THE LOWEST WEIGHT

Two reasons, one geological and one epistemic. Geologically it is the least specific evidence — it says the neighbourhood is right, not the ground. Epistemically it is **the layer most contaminated by exploration history**: known deposits are where people have already looked, so weighting it heavily would bias the model toward re-discovering the accessible. Keeping it at 0.10 limits that feedback loop.

CURRENT CONSTRAINT

The reference set is a curated seed of West African orogenic-gold deposits, so this layer contributes approximately nothing outside that region. Loading the full USGS MRDS is a known, scoped piece of work.

4 · Why these data sources

Every source had to clear four bars: global coverage, a licence permitting commercial use, stable programmatic access, and a resolution appropriate to the decision it informs.

SOURCE	WHY THIS ONE	WHAT IT FEEDS
Sentinel-2 L2A	10–20 m, free, global, ~5-day revisit, atmospherically corrected, and openly licensed for commercial use. The SWIR bands needed for hydroxyl and clay discrimination are the reason it is preferred over higher-resolution optical sources that lack them.	Iron oxide (B04/B02), clay and hydroxyl (B11/B12), NDVI as a vegetation mask
Copernicus DEM GLO-30	30 m global elevation with consistent quality and an open licence — unlike SRTM it has systematic void handling and better coverage at the latitudes that matter here.	Slope, hillshade, terrain context and access
NOAA EMAG2v3	The only global magnetic-anomaly compilation with an open licence. At roughly 2 arc-minutes it is far too coarse for target definition, which is exactly why it is used only as regional structural context and never as primary evidence.	Regional anomaly and analytic signal, as context layers
Macrostrat	Global, programmatically queryable geological map units with lithology strings — machine-readable geology, which most national survey products are not.	Lithology favourability (weight 0.15)
OneGeology / BRGM	Authoritative national and regional geological mapping for visual interpretation where Macrostrat's generalisation is too coarse.	Interpretation backdrop, not scored
USGS MRDS	The most complete openly licensed deposit register. Used for both proximity scoring and, separately, as the authority against which any deposit name in generated text is verified.	Deposit proximity (0.10); claim verification
WDPA protected areas	The recognised global reference for conservation boundaries, which is what a regulator or financier will check against.	Hard exclusion veto
OpenStreetMap	Better road coverage in West Africa than most commercial datasets, and openly licensed.	Access and logistics context

THE DELIBERATE ASYMMETRY

Public data is used for **context and weighting**; the user's own survey data is the primary anomaly evidence. Regional magnetics at 2 arc-

minutes cannot define a drill target and is never asked to. This ordering is why the platform is useful on day one but becomes materially better once a customer uploads their own ground or airborne survey.

5 · Processing decisions that change the answer

Five choices in the pipeline that are not obvious, each made for a reason that survives being asked about.

01 **Kriging with a bounded variogram, and a fallback chain**

Point survey data is interpolated by ordinary kriging. The variogram model is tried in the order **spherical, exponential, gaussian, linear**, because real geophysical surveys have a **bounded** semivariogram — semivariance rises with separation and then flattens at a sill, because geological bodies have finite size. A linear model has no sill and cannot fit that shape; least squares lands on near-zero slope with a huge nugget, meaning "no spatial correlation at any distance". Kriging then weights every station almost equally and predicts the global mean everywhere, producing an essentially constant grid. The pipeline detects that degeneracy explicitly and falls through to the next model.

02 **Percentile clipping, not min–max, when normalising**

Each evidence column is scaled to [0, 1] against its **1st and 99th percentiles**. Raw min–max is dominated by single extreme cells — a kriging edge artefact, a survey spike, a bad station — which compress every genuine value into a narrow band near zero. Sparse columns defeat percentile clipping too (binned geochemistry is over 99% zeros, so the 99th percentile is itself zero), so a degenerate clip range falls back to full min–max rather than silently flattening the layer.

03 **Missing data is unfavourable, not neutral**

A cell with no coverage for a layer scores zero on it. This is a deliberate conservative choice: it means **absence of evidence reduces confidence** rather than being waved through. It matters most on the inverted magnetic layer, where treating no-data as neutral would let a cell with no magnetic coverage read as a perfect magnetic low and surface as a target on the strength of missing data.

04 **Targets are clustered, not just top-ranked cells**

The highest-scoring cells are grouped with **HDBSCAN**, a density-based clustering method, before ranking. Without this, the top five cells are typically five adjacent cells on one anomaly — five "targets" that are one target. Density-based clustering is used rather than a fixed radius because anomalies are not a uniform size. Each resulting target reports its peak score, mean score and cell count, so a broad moderate anomaly can be told apart from a sharp intense one.

05 **Exclusions veto rather than down-weight**

Cells inside a protected area are **zeroed after scoring**, not fed in as another weighted layer. A protected area is a regulatory fact, not geological evidence, and no amount of favourability should let it be outvoted. Keeping the veto outside the weighted model is what makes that guarantee absolute.

6 · How the features map to disclosure practice

Several product decisions look like engineering choices and are actually compliance ones. This is the reasoning to give a Qualified Person.

FEATURE	THE STANDARD IT SERVES
Per-layer provenance with checksums	Reporting codes require disclosure of data source, quality and processing. Recording the source file, its checksum, every processing step and the algorithm parameters means the lineage is a stored record rather than a recollection.
SHAP attribution per target	A ranked target is an interpretation, and interpretations must be supportable. Per-target evidence contributions let a competent person state which data drove which conclusion, and disagree with it specifically.
Qualified-Person annotation and review status	Reporting codes require a named competent person to take responsibility. Notes, status and reviewer identity attach to the target and follow it into every export, so the sign-off travels with the claim.
Versioned reports	Regenerating creates a new version and preserves the previous one, so what was disclosed at a point in time remains reconstructable.
Model version recorded on results	If the model changes, prior results remain attributable to the model that produced them.
QP and investor personas strip unsupported claims	These two audiences carry the most liability. Their report voices are explicitly instructed to caveat, flag assumptions, and remove any claim not backed by a data fact.
Deposit names verified before display	Naming a nearby deposit as an analogue is a material claim. Every name is checked against a deposit register and removed if unmatched, so an analogue cannot be invented.
Ranking withheld for unmodelled commodities	Producing a number from a deposit model that does not apply is worse than producing none. The refusal is the compliance feature.

7 · Multi-commodity: what changes and what does not

The intention is explicit and the architecture already has the seam. What follows is the plan of record, not an aspiration.

Already global today. Everything except the ranking. Satellite indices, mapped lithology, terrain, magnetics and geochemistry ingestion, the map, coordinate handling and exports all work anywhere on Earth for any commodity. Only the scoring profile is gold-specific.

Three things are gold-specific. The weight set; the lithology favourability table; and the reference-deposit set. Nothing else.

The planned change replaces the single weight set with a profile registry keyed by commodity — each profile carrying its own weights, lithology favourability and notes. Scoring takes a profile, defaulting to orogenic gold so existing results are unchanged and regression-locked.

PLANNED PROFILE	EVIDENCE EMPHASIS	WHAT IT NEEDS THAT DOES NOT EXIST YET
Orogenic gold	Structure, Au/As geochemistry, magnetic low, greenstone host, deposit proximity	Shipped
Porphyry copper	Magnetic high from the causative intrusion, potassic alteration, Cu/Mo geochemistry, intrusive host, concentric structure	A signed magnetic feature — polarity is opposite to gold, so gradient magnitude alone is insufficient. ASTER alteration ratios.
Lithium pegmatite	Fractionated granite and pegmatite host, Li/Cs/Ta geochemistry, radiometric K/Th, structural corridors	Radiometrics as a new upload type and processing pipeline. An inverted lithology table.
Magmatic nickel–PGE	Mafic and ultramafic host, magnetic and electromagnetic conductors, Ni/Cu/PGE geochemistry	Electromagnetic and conductivity data as a new upload type.

THE HONEST SEQUENCING ARGUMENT

Each profile is an **expert prior in exactly the same transparent form**, refined later with real drill labels per commodity. The reason they are not built yet is not difficulty — the gold-parity refactor is roughly a day, and each additional profile half a day to a day where no new data pipeline is needed. It is that a **second unvalidated model is not progress**. Breadth before the first depth is how a prospectivity product becomes plausible everywhere and trusted nowhere.

8 · What a reviewer should press on

The questions a competent person ought to ask. Better to have the answers ready than to be surprised by them.

- **The magnetic polarity assumption.** Named as the weakest link, with three documented counter-cases and field validation still outstanding. If the host is non-magnetic or banded iron formation, this layer is at best uninformative.
- **Structure is inferred from magnetics alone.** Gradient magnitude is a proxy for structural discontinuity, not a mapped structure. It will respond to any magnetic contact, mineralised or not.
- **Deposit proximity encodes exploration history.** Held at the lowest weight for that reason, but it is not eliminated.
- **The expert prior is not yet calibrated against drill outcomes.** It is a defensible synthesis of published practice, not a validated predictor, and should be described that way.
- **Grid resolution bounds the claim.** Targets are cell-scale features on the analysis grid; they locate ground worth investigating, not a collar position.
- **Depth estimates are model-dependent.** Euler deconvolution requires an assumed structural index, and the answer moves with that assumption.

ONE ITEM TO CHECK BEFORE PRESENTING

The Euler depth solver uses **structural index 2** while its code comment describes the source as "contact/dyke-like". Under the conventional magnetic structural-index set, 0 is a contact, 1 a dyke or sill, and 2 a pipe or horizontal cylinder — so the value and the description may not agree. This does not affect scoring or ranking, only the reported depth estimate, but it is worth a geophysicist's eye before depth numbers are defended in a technical meeting.

